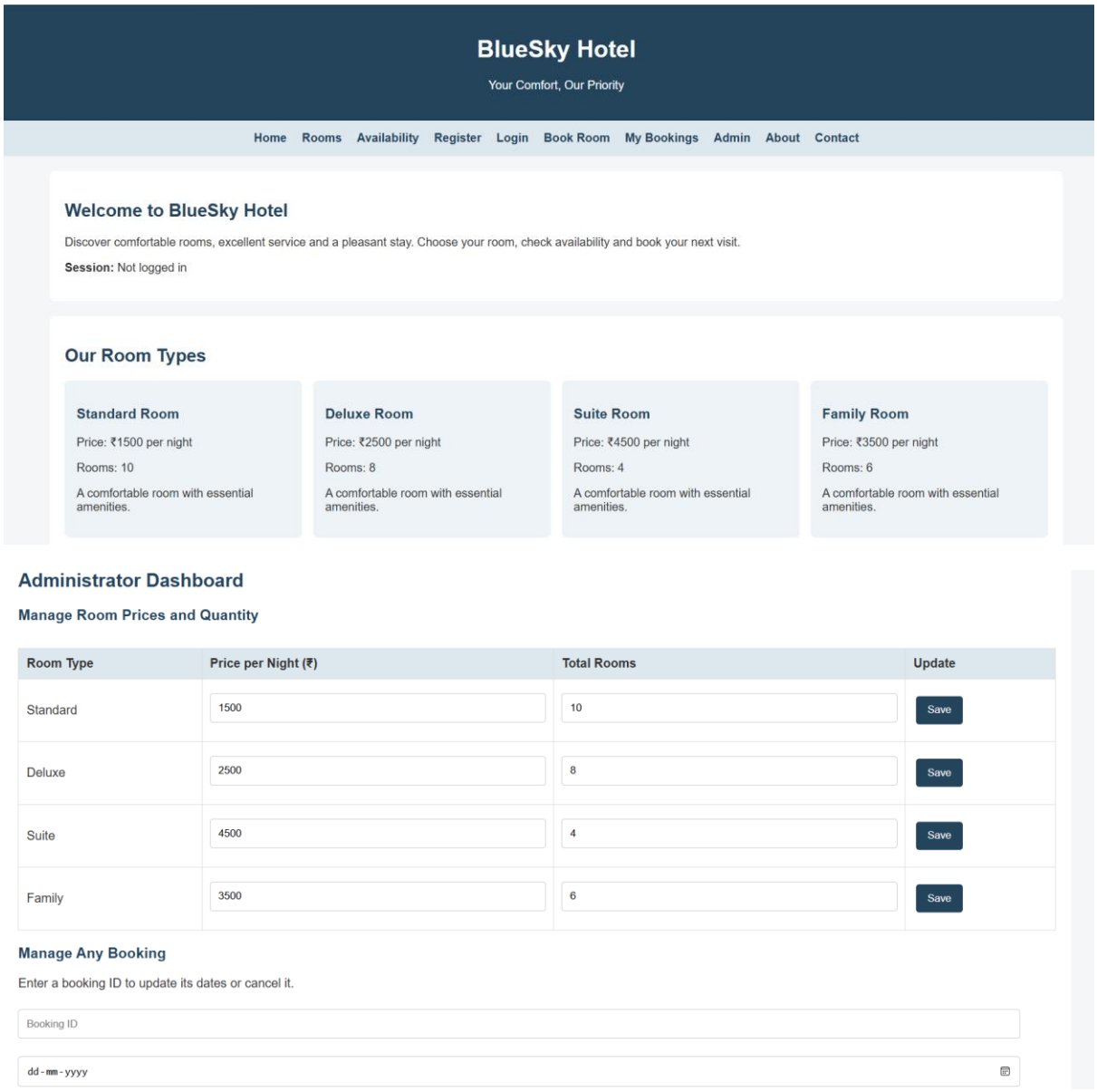
Exp2-Hotel Booking




```
<!DOCTYPE html>
<html>
<head>
  <title>Hotel Room Booking System</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <style>
```

```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  background: #f4f6f8;
  color: #333;
}

header {
  background: #24445c;
  color: white;
  text-align: center;
  padding: 22px;
}

nav {
  background: #dce7ee;
  padding: 15px;
  text-align: center;
}

nav a {
  color: #24445c;
  text-decoration: none;
  font-weight: bold;
  margin: 8px;
}

section {
```

```
  background: white;
  width: 88%;
  margin: 20px auto;
  padding: 20px;
  border-radius: 8px;
}

h2, h3 { color: #24445c; }

input, select, textarea {
  width: 95%;
  padding: 10px;
  margin: 8px 0 14px;
  border: 1px solid #ccc;
  border-radius: 4px;
}

button {
  background: #24445c;
  color: white;
  padding: 10px 15px;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  margin: 3px;
}

button:hover { background: #386781; }
```

```
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 12px;
}

th, td {
  padding: 10px;
  border: 1px solid #ddd;
  text-align: left;
}

th { background: #dce7ee; }

.rooms {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 15px;
}

.room {
  background: #edf3f7;
  padding: 16px;
  border-radius: 7px;
}

.hidden { display: none; }
```

```
    footer {
      background: #24445c;
      color: white;
      text-align: center;
      padding: 16px;
    }

    @media(max-width:600px) {
      section { width: 88%; }
      nav a { display: inline-block; }
      table { font-size: 12px; }
    }
  </style>
</head>

<body>

<header>
  <h1>BlueSky Hotel</h1>
  <p>Your Comfort, Our Priority</p>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#rooms">Rooms</a>
  <a href="#availability">Availability</a>
  <a href="#register">Register</a>
```

```
    <a href="#login">Login</a>
    <a href="#booking">Book Room</a>
    <a href="#history">My Bookings</a>
    <a href="#admin">Admin</a>
    <a href="#about">About</a>
    <a href="#contact">Contact</a>
</nav>

<section id="home">
    <h2>Welcome to BlueSky Hotel</h2>
    <p>
        Discover comfortable rooms, excellent service and a pleasant stay.
        Choose your room, check availability and book your next visit.
    </p>
    <p><b>Session:</b> <span id="status">Not logged in</span></p>
</section>

<section id="rooms">
    <h2>Our Room Types</h2>
    <div class="rooms" id="roomCards"></div>
</section>

<section id="availability">
    <h2>Check Room Availability</h2>
    <form onsubmit="checkAvailability(event)">
        <label>Check-in Date</label>
        <input type="date" id="checkIn" required>
```

```
      <label>Check-out Date</label>
      <input type="date" id="checkOut" required>

      <label>Number of Guests</label>
      <input type="number" id="guests" min="1" value="1" required>

      <label>Room Type</label>
      <select id="availableType">
        <option>Standard</option>
        <option>Deluxe</option>
        <option>Suite</option>
        <option>Family</option>
      </select>

      <button type="submit">Check Availability</button>
    </form>
    <p id="availabilityResult"></p>
</section>

<section id="register">
    <h2>User Registration</h2>
    <form onsubmit="registerUser(event)">
      <label>Full Name</label>
      <input id="regName" required>

      <label>Email</label>
      <input type="email" id="regEmail" required>
```

```
      <label>Password</label>
      <input type="password" id="regPassword" required>

      <button type="submit">Register</button>
    </form>
    <p id="regMessage"></p>
</section>

<section id="login">
    <h2>Login</h2>
    <form onsubmit="loginUser(event)">
      <label>Email</label>
      <input type="email" id="loginEmail" required>

      <label>Password</label>
      <input type="password" id="loginPassword" required>

      <label>Login As</label>
      <select id="loginRole">
        <option value="user">Customer</option>
        <option value="admin">Administrator</option>
      </select>

      <button type="submit">Login</button>
    </form>
    <p id="loginMessage"></p>
    <p>Demo Admin: admin@hotel.com / admin123</p>
    <p>Demo User: user@hotel.com / user123</p>
```

```
</section>

<section id="booking">
  <h2>Book a Room</h2>
  <form onsubmit="bookRoom(event)">
    <label>Room Type</label>
    <select id="bookType">
      <option>Standard</option>
      <option>Deluxe</option>
      <option>Suite</option>
      <option>Family</option>
    </select>

    <label>Check-in Date</label>
    <input type="date" id="bookIn" required>

    <label>Check-out Date</label>
    <input type="date" id="bookOut" required>

    <label>Number of Guests</label>
    <input type="number" id="bookGuests" min="1" value="1" required>

    <button type="submit">Confirm Booking</button>
  </form>
  <p id="bookingMessage"></p>
  <p class="small">Payment gateway and confirmation emails are not connected in this demo.</p>
</section>
```

```
<section id="history">
  <h2>My Booking History</h2>
  <div style="overflow-x:auto">
    <table>
      <thead>
        <tr>
          <th>Booking ID</th>
          <th>Room</th>
          <th>Check-in</th>
          <th>Check-out</th>
          <th>Guests</th>
          <th>Total</th>
          <th>Status</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody id="bookingHistory"></tbody>
    </table>
  </div>

  <h3>Update Profile</h3>
  <form onsubmit="updateProfile(event)">
    <input id="profileName" placeholder="Full Name" required>
    <input id="profileEmail" type="email" placeholder="Email" disabled>
    <button type="submit">Update Profile</button>
  </form>
```

```
  <h3>Rate Your Stay</h3>
  <form onsubmit="addReview(event)">
    <label>Rating</label>
    <select id="rating">
      <option value="5">5 - Excellent</option>
      <option value="4">4 - Very Good</option>
      <option value="3">3 - Good</option>
      <option value="2">2 - Fair</option>
      <option value="1">1 - Poor</option>
    </select>
    <textarea id="reviewText" placeholder="Write your review" required></textarea>
    <button type="submit">Submit Review</button>
  </form>
  <p id="reviewMessage"></p>
</section>

<section id="admin">
  <h2>Administrator Dashboard</h2>

  <h3>Manage Room Prices and Quantity</h3>
  <div style="overflow-x:auto">
    <table>
      <thead>
        <tr>
          <th>Room Type</th>
          <th>Price per Night (₹)</th>
          <th>Total Rooms</th>
          <th>Update</th>
```

```
        </tr>
      </thead>
      <tbody id="roomAdmin"></tbody>
    </table>
  </div>

  <h3>Manage Any Booking</h3>
  <p>Enter a booking ID to update its dates or cancel it.</p>
  <input type="number" id="adminBookingId" placeholder="Booking ID">
  <input type="date" id="adminIn">
  <input type="date" id="adminOut">
  <button onclick="adminUpdateBooking()">Update Booking</button>
  <button onclick="adminCancelBooking()">Cancel Booking</button>

  <h3>All Customer Bookings</h3>
  <div style="overflow-x:auto">
    <table>
      <thead>
        <tr>
          <th>ID</th>
          <th>Customer</th>
          <th>Room</th>
          <th>Dates</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody id="allBookings"></tbody>
    </table>
```

```
  </div>
</section>

<section id="about">
  <h2>About Us</h2>
  <p id="aboutText">
    BlueSky Hotel provides comfortable accommodation, quality hospitality
    and convenient booking services for families and travellers.
  </p>

  <h2>Customer Service - FAQs</h2>
  <h3>What time is check-in?</h3>
  <p>Standard check-in is at 12:00 PM.</p>
  <h3>Can I cancel my booking?</h3>
  <p>Bookings can be cancelled at least three days before check-in.</p>
  <h3>How can I modify my booking?</h3>
  <p>Log in and update your booking at least three days before arrival.</p>

  <h2>Local Travel Guide</h2>
  <p>Ask our reception team about nearby attractions, public transport,
    local restaurants and travel options.</p>

  <h2>Privacy Policy</h2>
  <p>Customer information should be collected only for booking and service
    purposes. A production website must provide secure storage and
    appropriate privacy protections.</p>

  <h3>Administrator: Edit About Information</h3>
```

```
  <textarea id="aboutEdit" rows="3" placeholder="Enter new About Us information"></textarea>
  <button onclick="updateAbout()">Update About Us</button>
</section>

<section id="contact">
  <h2>Contact Us</h2>
  <p>Email: contact@blueskyhotel.example</p>
  <p>Phone: +91 98765 43210</p>
  <p>Address: Main Road, City Centre</p>

  <form onsubmit="sendContact(event)">
    <input id="contactName" placeholder="Your Name" required>
    <input type="email" id="contactEmail" placeholder="Your Email" required>
    <textarea id="contactMessage" placeholder="Your Message" required></textarea>
    <button type="submit">Send Message</button>
  </form>
  <p id="contactResult"></p>

  <h3>Administrator: Edit Contact Details</h3>
  <textarea id="contactEdit" rows="3" placeholder="Enter updated contact information"></textarea>
  <button onclick="updateContact()">Update Contact</button>
</section>

<footer>
  <p>BlueSky Hotel | Hotel Room Booking System</p>
  <p>&copy; 2026 All Rights Reserved</p>
</footer>
```

```
<script>
// All data is kept in memory for this session only.
let rooms = [
  {type:"Standard", price:1500, quantity:10},
  {type:"Deluxe", price:2500, quantity:8},
  {type:"Suite", price:4500, quantity:4},
  {type:"Family", price:3500, quantity:6}
];

let users = [
  {name:"Demo User", email:"user@hotel.com", password:"user123"}
];

let bookings = [];
let reviews = [];
let currentUser = null;
let nextId = 1;

function validDates(start, end) {
  return start >= new Date().toISOString().slice(0,10) && end > start;
}

function nights(start, end) {
  return Math.round((new Date(end) - new Date(start)) / 86400000);
}

function roomPrice(type) {
```

```
  return rooms.find(r => r.type === type).price;
}

function roomAvailable(type, start, end, ignoreId = -1) {
  let room = rooms.find(r => r.type === type);
  let overlapping = bookings.filter(b =>
    b.type === type && b.status === "Confirmed" && b.id !== ignoreId &&
    start < b.out && end > b.in
  ).length;

  return room && overlapping < room.quantity;
}

function loginUser(event) {
  event.preventDefault();

  let email = document.getElementById("loginEmail").value.toLowerCase();
  let password = document.getElementById("loginPassword").value;
  let role = document.getElementById("loginRole").value;

  if (role === "admin" && email === "admin@hotel.com" && password === "admin123") {
    currentUser = {name:"Administrator", email:email, role:"admin"};
  } else {
    let user = users.find(u => u.email.toLowerCase() === email && u.password ===
password);

    if (role !== "user" || !user) {
      document.getElementById("loginMessage").textContent = "Invalid login details.";
```

```
        return;
      }

      currentUser = {...user, role:"user"};
    }

    document.getElementById("loginMessage").textContent = "Login successful!";
    document.getElementById("status").textContent = currentUser.name + " (" +
currentUser.role + ")";
    refresh();
}

function registerUser(event) {
    event.preventDefault();

    let email = document.getElementById("regEmail").value.toLowerCase();

    if (users.some(u => u.email.toLowerCase() === email) || email ===
"admin@hotel.com") {
      document.getElementById("regMessage").textContent = "Email already registered.";
      return;
    }

    users.push({
      name:document.getElementById("regName").value,
      email:email,
      password:document.getElementById("regPassword").value
    });
```

```
  document.getElementById("regMessage").textContent = "Registration successful. Please log in.";
  event.target.reset();
  refresh();
}

function logout() {
  currentUser = null;
  document.getElementById("status").textContent = "Not logged in";
  document.getElementById("loginMessage").textContent = "You have logged out.";
  refresh();
}

function checkAvailability(event) {
  event.preventDefault();

  let type = document.getElementById("availableType").value;
  let start = document.getElementById("checkIn").value;
  let end = document.getElementById("checkOut").value;

  if (!validDates(start, end)) {
    document.getElementById("availabilityResult").textContent = "Select valid future dates.";
    return;
  }

  let available = roomAvailable(type, start, end);

  document.getElementById("availabilityResult").textContent =
```

```
    available ? type + " rooms are available. Price: ₹" + roomPrice(type) + " per night."
              : "No " + type + " rooms are available for these dates.";
}

function bookRoom(event) {
  event.preventDefault();

  if (!currentUser || currentUser.role !== "user") {
    alert("Please log in as a customer before booking.");
    return;
  }

  let type = document.getElementById("bookType").value;
  let start = document.getElementById("bookIn").value;
  let end = document.getElementById("bookOut").value;
  let guests = Number(document.getElementById("bookGuests").value);

  if (!validDates(start, end) || guests < 1) {
    alert("Please enter valid dates and guest details.");
    return;
  }

  if (!roomAvailable(type, start, end)) {
    alert("Sorry, this room type is unavailable for these dates.");
    return;
  }

  let total = roomPrice(type) * nights(start, end);
```

```
  bookings.push({
    id:nextId++,
    email:currentUser.email,
    type:type,
    in:start,
    out:end,
    guests:guests,
    total:total,
    status:"Confirmed"
  });

  document.getElementById("bookingMessage").textContent =
    "Booking confirmed! Total amount: ₹" + total;

  refresh();
}

function cancelBooking(id) {
  let b = bookings.find(b => b.id === id);

  if (!b || b.status !== "Confirmed") return;

  let deadline = new Date(b.in + "T00:00:00");
  deadline.setDate(deadline.getDate() - 3);

  if (new Date() > deadline) {
    alert("Bookings can only be cancelled at least 3 days before arrival.");
```

```
    return;
  }

  b.status = "Cancelled";
  refresh();
}

function changeBooking(id) {
  let b = bookings.find(b => b.id === id);

  if (!b || b.status !== "Confirmed") return;

  let deadline = new Date(b.in + "T00:00:00");
  deadline.setDate(deadline.getDate() - 3);

  if (new Date() > deadline) {
    alert("Bookings can only be changed at least 3 days before arrival.");
    return;
  }

  let start = prompt("New check-in date (YYYY-MM-DD):", b.in);
  if (!start) return;

  let end = prompt("New check-out date (YYYY-MM-DD):", b.out);
  if (!end) return;

  if (!validDates(start, end) || !roomAvailable(b.type, start, end, b.id)) {
    alert("Invalid dates or no rooms available.");
```

```
      return;
    }

    b.in = start;
    b.out = end;
    b.total = roomPrice(b.type) * nights(start, end);
    refresh();
    alert("Booking updated.");
}

function updateProfile(event) {
    event.preventDefault();

    if (!currentUser || currentUser.role !== "user") {
      alert("Please log in as a customer.");
      return;
    }

    let user = users.find(u => u.email === currentUser.email);
    user.name = document.getElementById("profileName").value;
    currentUser.name = user.name;

    document.getElementById("status").textContent = user.name + " (user)";
    alert("Profile updated.");
}

function addReview(event) {
    event.preventDefault();
```

```
  if (!currentUser || currentUser.role !== "user") {
    alert("Please log in first.");
    return;
  }

  reviews.push({
    name:currentUser.name,
    rating:document.getElementById("rating").value,
    text:document.getElementById("reviewText").value
  });

  document.getElementById("reviewMessage").textContent = "Thank you for your
review!";
  event.target.reset();
}

function updateRoom(index) {
  let price = Number(document.getElementById("price" + index).value);
  let quantity = Number(document.getElementById("quantity" + index).value);

  if (price <= 0 || quantity < 0 || !Number.isInteger(quantity)) {
    alert("Enter a positive price and a valid room quantity.");
    return;
  }

  let confirmed = bookings.filter(b =>
    b.type === rooms[index].type && b.status === "Confirmed"
```

```
    ).length;

    if (quantity < confirmed) {
      alert("Room quantity cannot be less than confirmed bookings.");
      return;
    }

    rooms[index].price = price;
    rooms[index].quantity = quantity;
    refresh();
    alert("Room details updated.");
}

function adminUpdateBooking() {
    if (!currentUser || currentUser.role !== "admin") {
      alert("Admin login required.");
      return;
    }

    let id = Number(document.getElementById("adminBookingId").value);
    let b = bookings.find(b => b.id === id && b.status === "Confirmed");

    if (!b) {
      alert("Confirmed booking not found.");
      return;
    }

    let start = document.getElementById("adminIn").value;
```

```
  let end = document.getElementById("adminOut").value;

  if (!validDates(start, end) || !roomAvailable(b.type, start, end, b.id)) {
    alert("Invalid dates or room unavailable.");
    return;
  }

  b.in = start;
  b.out = end;
  b.total = roomPrice(b.type) * nights(start, end);
  refresh();
  alert("Booking updated.");
}

function adminCancelBooking() {
  if (!currentUser || currentUser.role !== "admin") {
    alert("Admin login required.");
    return;
  }

  let id = Number(document.getElementById("adminBookingId").value);
  let b = bookings.find(b => b.id === id && b.status === "Confirmed");

  if (!b) {
    alert("Confirmed booking not found.");
    return;
  }
```

```
  b.status = "Cancelled";
  refresh();
  alert("Booking cancelled by admin.");
}

function updateAbout() {
  if (!currentUser || currentUser.role !== "admin") {
    alert("Admin login required.");
    return;
  }

  let text = document.getElementById("aboutEdit").value.trim();
  if (text) document.getElementById("aboutText").textContent = text;
}

function updateContact() {
  if (!currentUser || currentUser.role !== "admin") {
    alert("Admin login required.");
    return;
  }

  alert("Contact details edited for demonstration. Connect a backend to save them permanently.");
}

function sendContact(event) {
  event.preventDefault();
  document.getElementById("contactResult").textContent =
```

```
    "Message submitted for demonstration. No email was sent.";
  event.target.reset();
}

function refresh() {
  document.getElementById("roomCards").innerHTML = rooms.map(r => `
    <div class="room">
      <h3>${r.type} Room</h3>
      <p>Price: ₹${r.price} per night</p>
      <p>Rooms: ${r.quantity}</p>
      <p>A comfortable room with essential amenities.</p>
    </div>
  `).join("");

  document.getElementById("roomAdmin").innerHTML = rooms.map((r,i) => `
    <tr>
      <td>${r.type}</td>
      <td><input id="price${i}" type="number" min="1" value="${r.price}"></td>
      <td><input id="quantity${i}" type="number" min="0" value="${r.quantity}"></td>
      <td><button onclick="updateRoom(${i})">Save</button></td>
    </tr>
  `).join("");

  let myBookings = currentUser && currentUser.role === "user"
    ? bookings.filter(b => b.email === currentUser.email) : [];

  document.getElementById("bookingHistory").innerHTML = myBookings.map(b => `
    <tr>
```

```
      <td>${b.id}</td>
      <td>${b.type}</td>
      <td>${b.in}</td>
      <td>${b.out}</td>
      <td>${b.guests}</td>
      <td>₹${b.total}</td>
      <td>${b.status}</td>
      <td>${b.status === "Confirmed" ? `
        <button onclick="changeBooking(${b.id})">Change</button>
        <button onclick="cancelBooking(${b.id})">Cancel</button>
      ` : "—"}</td>
    </tr>
  `).join("") || "<tr><td colspan='8'>No bookings yet.</td></tr>";

  document.getElementById("allBookings").innerHTML = bookings.map(b => `
    <tr>
      <td>${b.id}</td>
      <td>${b.email}</td>
      <td>${b.type}</td>
      <td>${b.in} to ${b.out}</td>
      <td>${b.status}</td>
    </tr>
  `).join("") || "<tr><td colspan='5'>No bookings yet.</td></tr>";

  if (currentUser && currentUser.role === "user") {
    document.getElementById("profileName").value = currentUser.name;
    document.getElementById("profileEmail").value = currentUser.email;
  }
```

```
}

// Initial page rendering
refresh();
</script>

</body>
</html>
```